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System and method for converting helium bath cooling system for a superconducting magnet of a magnetic resonance imaging system into a sealed cryogenic system

Abstract

A superconducting magnet system includes a coil support structure having a body, a superconducting magnet having a plurality of coils disposed about the body, and a helium vessel encompassing the coil support structure and the superconducting magnet. The superconducting magnet system includes a cooling system that includes a network of tubes disposed within the helium vessel and configured to transport helium within, wherein the network of tubes is retroactively thermally coupled to the plurality of coils. The cooling system includes a liquid helium storage system located within the helium vessel and a helium recondenser located outside the helium vessel. The cooling system includes one or more tubes coupling the helium storage system to the helium recondenser, wherein where the one or more tubes pass through the helium vessel is retroactively vacuum sealed to make the helium vessel a sealed helium vessel.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2009/0293980	12/2008	Burke, II	138/146	F16L 9/02
2011/0160064	12/2010	Pfleiderer	165/185	F25D 19/006
2014/0274721	12/2013	Calvert	324/309	G01R 33/3854
2016/0187435	12/2015	Jiang	62/51.1	G01R 33/3804
2016/0187439	12/2015	Lvovsky	324/309	G01R 33/543
2017/0250018	12/2016	Xu	N/A	H01F 6/04
2018/0144852	12/2017	Ito	N/A	F25B 9/14

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2022068465	12/2021	WO	N/A

OTHER PUBLICATIONS

U.S. Appl. No. 17/969,024, filed Oct. 19, 2022, Celik. cited by applicant

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Background/Summary

BACKGROUND

(1) The subject matter disclosed herein relates to a superconducting magnet for a magnetic resonance imaging (MRI) and, more particularly, a system and a method for converting helium bath cooling system for a superconducting magnet of an MRI system into a sealed cryogenic system.

(2) Non-invasive imaging technologies allow images of the internal structures or features of a patient/object to be obtained without performing an invasive procedure on the patient/object. In particular, such non-invasive imaging technologies rely on various physical principles (such as the

differential transmission of X-rays through a target volume, the reflection of acoustic waves within the volume, the paramagnetic properties of different tissues and materials within the volume, the breakdown of targeted radionuclides within the body, and so forth) to acquire data and to construct images or otherwise represent the observed internal features of the patient/object.

(3) During MRI, when a substance such as human tissue is subjected to a uniform magnetic field (polarizing field $B_{sub.0}$), the individual magnetic moments of the spins in the tissue attempt to align with this polarizing field, but precess about it in random order at their characteristic Larmor frequency. If the substance, or tissue, is subjected to a magnetic field (excitation field $B_{sub.1}$) which is in the x-y plane and which is near the Larmor frequency, the net aligned moment, or “longitudinal magnetization”, $M_{sub.z}$, may be rotated, or “tipped”, into the x-y plane to produce a net transverse magnetic moment, $M_{sub.t}$. A signal is emitted by the excited spins after the excitation signal $B_{sub.1}$ is terminated and this signal may be received and processed to form an image.

(4) When utilizing these signals to produce images, magnetic field gradients ($G_{sub.x}$, $G_{sub.y}$, and $G_{sub.z}$) are employed. Typically, the region to be imaged is scanned by a sequence of measurement cycles in which these gradient fields vary according to the particular localization method being used. The resulting set of received nuclear magnetic resonance (NMR) signals are digitized and processed to reconstruct the image using one of many well-known reconstruction techniques.

(5) MRI systems may utilize superconducting magnets. Also, MRI systems may utilize liquid helium to operate. Sealed cryogenic systems utilize very small amounts of helium (e.g., a few liters versus several hundreds or even a couple of thousand liters utilized by bath cooling the superconducting magnet). All existing sealed cryogenic magnets use metal coil support structures (e.g., formers) that take advantage of higher thermal conductivity for temperature uniformity. Also, metal support structures make thermal management easier. Further, cartridge/magnet suspension is much easier with metal formers. Liquid helium may also be utilized in bath cooling the superconducting magnet (e.g., by filling a cryogenic vessel with the superconducting magnet disposed inside (e.g. helium vessel) with liquid helium). In contrast to sealed cryogenic magnets, bath cooled magnets may utilize fiberglass composite structures which present many challenges. For example, the fiberglass composite structures have very low thermal conductivity that can create temperature gradients that eventually can cause a quench, larger outgassing, and emissivity. The larger outgassing and emissivity cause additional heat loads, which make it difficult to be part of a sealed system. Further, suspension attachment to the fiberglass composite support structures is challenging and may cause distortions in the fiberglass composite coil support structure and magnet homogeneity issues. Helium is a commodity whose price has been increasing due to global supply chain issues and increased demand. While there is a need to increase an efficiency of helium usage for a bath cooled magnet, it would be difficult due to issues with the fiberglass composite coil support structures.

SUMMARY

(6) A summary of certain embodiments disclosed herein is set forth below. It should be understood that these aspects are presented merely to provide the reader with a brief summary of these certain embodiments and that these aspects are not intended to limit the scope of this disclosure. Indeed, this disclosure may encompass a variety of aspects that may not be set forth below.

(7) In one embodiment, a superconducting magnet system for a magnetic resonance imaging system is provided. The superconducting magnet system a coil support structure having a body. The superconducting magnet system also includes a superconducting magnet having a plurality of coils disposed about the body of the coil support structure. The superconducting magnet system further includes a helium vessel encompassing the coil support structure and the superconducting magnet. The superconducting magnet system further includes a cooling system. The cooling system includes a network of tubes disposed within the helium vessel and configured to transport helium

within, wherein the network of tubes is retroactively thermally coupled to the plurality of coils. The cooling system also includes a liquid helium storage system located within the helium vessel. The cooling system further includes a helium recondenser located outside the helium vessel to recondense vapor helium to liquid helium. The cooling system further includes one or more tubes coupling the helium storage system to the helium recondenser, wherein where the one or more tubes pass through the helium vessel is retroactively vacuum sealed to make the helium vessel a sealed helium vessel, wherein the cooling system is configured to keep a uniform temperature between the sealed helium vessel and the plurality of coils.

(8) In another embodiment, a method for cooling a superconducting magnet for a magnetic resonance imaging system is provided. The method includes filling a sealed cryogenic vessel with liquid helium until a first desired temperature is reached for cooling the superconducting magnet having a plurality of coils disposed about a body of a coil support structure, wherein the superconducting magnet is disposed within the sealed cryogenic vessel. The method also includes ceasing filling the sealed cryogenic vessel with the liquid helium upon reaching the first desired temperature. The method further includes utilizing a roughing vacuum pump coupled to the sealed cryogenic vessel to evacuate excess helium gas after ceasing filling the sealed cryogenic vessel with the liquid helium. The method even further includes activating a sealed cryogenic system including a network of tubes disposed within the sealed cryogenic vessel by flowing a cryogen through the network of tubes to cool the superconducting magnet to a second desired temperature, wherein the second desired temperature is lower than the first desired temperature.

(9) In a further embodiment, a magnetic resonance imaging system is provided. The magnetic resonance imaging system includes a coil support structure having a body. The body of the coil support structure is made of fiberglass composite. The magnetic resonance imaging system also includes a superconducting magnet having a plurality of coils disposed about the body of the coil support structure. The magnetic resonance imaging system further includes a helium vessel encompassing the coil support structure and the superconducting magnet. The magnetic resonance imaging system also includes a cooling system. The cooling system includes a network of tubes disposed within the helium vessel and configured to transport helium within, wherein the network of tubes is retroactively thermally coupled to the plurality of coils. The cooling system further includes a liquid helium storage system located within the helium vessel. The cooling system further includes a helium recondenser located outside the helium vessel to recondense vapor helium to liquid helium. The cooling system even further includes one or more tubes coupling the helium storage system to the helium recondenser, wherein where the one or more tubes pass through the helium vessel is retroactively vacuum sealed to make the helium vessel a sealed helium vessel. The cooling system is configured to initially cool the superconducting magnet to a first desired temperature via filling the sealed helium vessel with liquid helium and to subsequently cool the superconducting magnet to a second desired temperature via the helium flowing through the network of tubes, wherein the second desired temperature is lower than the first desired temperature.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) These and other features, aspects, and advantages of the present disclosure will become better understood when the following detailed description is read with reference to the accompanying drawings in which like characters represent like parts throughout the drawings, wherein:

(2) FIG. 1 is a side cross-sectional view of a portion of cooling system for a magnet resonance imaging machine, in accordance with aspects of the present disclosure;

(3) FIG. 2 is a schematic diagram of a cooling system for a magnet resonance imaging machine

- (including the portion of the cooling system in FIG. 1), in accordance with aspects of the present disclosure;
- (4) FIG. 3 is a cross-sectional view of a cooling tube of the cooling system in FIG. 1, in accordance with aspects of the present disclosure;
- (5) FIG. 4 is a schematic diagram illustrating a process for thermocoupling a coil of a superconducting magnet to the cooling tube in FIG. 3 (e.g., using metal wire overwrap), in accordance with aspects of the present disclosure;
- (6) FIG. 5 is a schematic diagram illustrating a process for thermocoupling a coil of a superconducting magnet to the cooling tube in FIG. 3 (e.g., using conforming metal plate), in accordance with aspects of the present disclosure;
- (7) FIG. 6 is a cross-sectional view of a cooling tube thermally coupled directly to the coil in FIG. 5, in accordance with aspects of the present disclosure;
- (8) FIG. 7 is a liquid helium network for cooling a superconducting magnet of an MRI machine, in accordance with aspects of the present disclosure;
- (9) FIG. 8 is a schematic diagram of a vacuum feedthrough between a recondenser and a tank of the cooling system in FIG. 2 (e.g., bellows), in accordance with aspects of the present disclosure;
- (10) FIG. 9 is a schematic diagram of a vacuum feedthrough between a recondenser and a tank of the cooling system in FIG. 2 (e.g., bi-metallic interface), in accordance with aspects of the present disclosure;
- (11) FIG. 10 is a schematic diagram of a bi-metallic interface coupled to a tube, in accordance with aspects of the present disclosure;
- (12) FIG. 11 is a schematic diagram of a superconducting switch, in accordance with aspects of the present disclosure;
- (13) FIG. 12 is a schematic diagram of a current leads coupled to a superconducting switch, in accordance with aspects of the present disclosure;
- (14) FIG. 13 is a flowchart of a method for cooling a superconducting magnet of a magnetic imaging system, in accordance with aspects of the present disclosure; and
- (15) FIG. 14 is a schematic diagram of an example magnetic resonance system, in accordance with aspects of the present disclosure.

DETAILED DESCRIPTION

(16) One or more specific embodiments will be described below. In an effort to provide a concise description of these embodiments, not all features of an actual implementation are described in the specification. It should be appreciated that in the development of any such actual implementation, as in any engineering or design project, numerous implementation-specific decisions must be made to achieve the developers' specific goals, such as compliance with system-related and business-related constraints, which may vary from one implementation to another. Moreover, it should be appreciated that such a development effort might be complex and time consuming, but would nevertheless be a routine undertaking of design, fabrication, and manufacture for those of ordinary skill having the benefit of this disclosure.

(17) When introducing elements of various embodiments of the present subject matter, the articles “a,” “an,” “the,” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. Furthermore, any numerical examples in the following discussion are intended to be non-limiting, and thus additional numerical values, ranges, and percentages are within the scope of the disclosed embodiments.

(18) While aspects of the following discussion are provided in the context of medical imaging, it should be appreciated that the disclosed techniques are not limited to such medical contexts. Indeed, the provision of examples and explanations in such a medical context is only to facilitate explanation by providing instances of real-world implementations and applications. However, the disclosed techniques may also be utilized in other contexts, such as image reconstruction for non-

destructive inspection of manufactured parts or goods (i.e., quality control or quality review applications), and/or the non-invasive inspection of packages, boxes, luggage, and so forth (i.e., security or screening applications). In general, the disclosed techniques may be useful in any imaging or screening context or image processing or photography field where a set or type of acquired data undergoes a reconstruction process to generate an image or volume.

(19) The present disclosure provides a system and a method for converting helium bath cooling system for a superconducting magnet of an MRI system into a sealed low cryogenic system. In particular, a superconducting magnet system includes a coil support structure (e.g., former) having a body. In certain embodiments, the coil support structure may be made of a fiberglass composite. The superconducting magnet system also includes a superconducting magnet having a plurality of coils disposed about the body of the coil support structure. The superconducting magnet system further includes a helium vessel encompassing the coil support structure and the superconducting magnet. The superconducting magnet system further includes a cooling system. The cooling system includes a network of tubes disposed within the helium vessel and configured to transport helium within, wherein the network of tubes is retroactively thermally coupled to the plurality of coils. The cooling system also includes a liquid helium storage system located within the helium vessel. The cooling system further includes a helium recondenser located outside the helium vessel to recondense vapor helium to liquid helium. The cooling system further includes one or more tubes coupling the helium storage system to the helium recondenser, wherein where the one or more tubes pass through the helium vessel is retroactively vacuum sealed to make the helium vessel a sealed helium vessel, wherein the cooling system is configured to keep a uniform temperature between the sealed helium vessel and the plurality of coils.

(20) In certain embodiments, in order to create the sealed helium vessel, the superconducting magnet system includes a respective bellows located where each respective tube of the one or more tubes pass through the helium vessel, wherein each respective bellows forms a vacuum seal and is coupled to the respective tube of the one or more tubes, and the one or more tubes are rigid. In certain embodiments, in order to create the sealed helium vessel, a respective bi-metallic interface disposed about each tube of the one or more tubes where each respective tube of the one or more tubes passes through the helium vessel, wherein each respective bi-metallic interface forms a vacuum seal, and the one or more tubes are flexible. In certain embodiments, each respective bi-metallic interface includes an inner metal layer that interfaces with the respective tube and an outer metal layer that interfaces with the helium vessel, and the inner metal layer is made of a different metallic material than the outer metal layer.

(21) In certain embodiments, each coil of the plurality of coils is thermally coupled to a respective tube of the network of tubes via a thermocoupling assembly, wherein the thermocoupling assembly includes a flat thermoconductive metal strap (e.g., flat copper strap) disposed on a respective coil, a thermoconductive metallic wire overwrap (e.g., copper wire overwrap) wound about both a portion of the flat thermoconductive metallic strap and the respective coil, and ends of the flat thermoconductive metal strap are coupled to a thermoconductive metallic plate (e.g., copper plate) coupled to the respective tube. In certain embodiments, the thermocoupling assembly includes a thermoconductive metallic sheet (e.g., copper sheet) wrapped around the respective tube, and the thermoconductive metallic plate is coupled to the thermoconductive metallic sheet. In certain embodiments, each coil of the plurality of coils is thermally coupled to a respective tube of the network of tubes via a thermocoupling assembly, wherein the thermocoupling assembly includes a flat thermoconductive metallic strap (e.g., flat copper strap) disposed on a respective coil, a thermoconductive metallic plate (e.g., copper plate) disposed about a portion of the flat thermoconductive metallic strap, and ends of the flat thermoconductive metallic strap are coupled to an additional thermoconductive metallic plate (e.g., copper plate) coupled to the respective tube. In certain embodiments, wherein the thermocoupling assembly includes a thermoconductive metallic sheet wrapped around the respective tube, and the additional thermoconductive metallic

plate is coupled to the thermoconductive metallic sheet.

(22) In certain embodiments, the superconducting magnet system includes a superconducting switch configured to operate at a temperature of 80 Kelvin or less, wherein the superconducting switch is both located within and thermally isolated from the helium vessel, the superconducting switch is configured to switch between a resistive mode and a superconducting mode, and the superconducting switch is thermally coupled to a respective tube of the network of tubes. In certain embodiments, the superconducting switch includes a blanket insulation disposed about a body of the switch and configured to thermally isolate the superconducting switch from the helium vessel, the superconducting switch includes a thermoconductive metallic sheet (e.g., copper sheet) disposed about the blanket insulation and thermally connected to the respective tube, and wherein the blanket insulation and the thermoconductive metallic sheet are configured to keep convection from being generated by the superconducting switch during ramping of the superconducting magnet. In certain embodiments, power leads coupled to the superconducting switch, wherein the power leads comprise a high temperature superconducting wire coupled to a heat sink located outside the helium vessel, wherein an end of the high temperature superconducting wire coupled to the heat sink is electrically isolated, and wherein the power leads comprise a low temperature superconducting wire having a first end coupled to the heat sink and a second end coupled to the superconducting switch, and the low temperature superconducting wire passes from outside the helium vessel through an electrically isolated feedthrough into the helium vessel.

(23) With the modifications noted above, the superconducting magnet can be cooled utilizing a combination of a cooling circuit thermoconductively directly coupled to the coils and helium bath cooling. A method for cooling a superconducting magnet for a magnetic resonance imaging system includes filling a sealed cryogenic vessel with liquid helium until a first desired temperature is reached for cooling the superconducting magnet having a plurality of coils disposed about a body of a coil support structure, wherein the superconducting magnet is disposed within the sealed cryogenic vessel. The method also includes ceasing filling the sealed cryogenic vessel with the liquid helium upon reaching the first desired temperature. The method further includes utilizing a roughing vacuum pump coupled to the sealed cryogenic vessel to evacuate excess helium gas after ceasing filling the sealed cryogenic vessel with the liquid helium. The method even further includes activating a sealed cryogenic system including a network of tubes disposed within the sealed cryogenic vessel by flowing a cryogen (e.g., helium) through the network of tubes to cool the superconducting magnet to a second desired temperature, wherein the second desired temperature is lower than the first desired temperature. In certain embodiments, the method includes filling the tube network with the cryogen and sealing it prior to cooling down the superconducting magnet. The pressure in the sealed cryogenic vessel after evacuating the excess helium gas generates internal convection to ensure a uniform temperature distribution between the plurality of coils and the sealed cryogenic vessel.

(24) The disclosed embodiments provide for converting existing bath cooled magnets (especially those with fiberglass composite structures) to sealed low cryogen systems while keeping the support structure, coil configuration, and cryostat (e.g., helium vessel) nearly intact, while replacing a main switch of the superconducting magnet with a switch configured for low cryogen operation and adding superconducting current leads. In addition, the disclosed embodiments provide for a liquid helium network that allows the cooling of the sealed helium vessel directly. Further, the disclosed embodiments provide for feedthrough configurations (e.g., for tubes through the sealed helium vessel) that eliminate vibration transmission from the coldhead and that compensate for thermal expansion differentials between the materials involved. Even further, the disclosed embodiments enable a faster cooldown for the superconducting magnet disposed in the sealed cryogenic vessel (e.g., from 1 to 3 weeks to 1 to 3 days) which is equivalent to cooldown for a bath cooled magnet. Still further, the disclosed embodiments reduces helium consumption (e.g., due to eliminating helium addition during a lifetime of the magnet) and, thus, the costs associated

with consumption of helium. Yet further, the disclosed embodiments eliminate the need for a quench vent pipe.

(25) Referring now to FIG. 1, a portion of a cooling system **10** for a superconducting magnet of a MRI machine is illustrated. The portion of the cooling system **10** is retroactively fitted to the MRI machine originally configured for utilizing helium bath cooling of the superconducting magnet. As shown therein, the cooling system **10** includes a plurality of cooling tubes **12** (e.g., a network of tubes **12** forming a sealed cryogenic system), or other suitable cooling paths, with liquid helium circulating within the cooling tubes **12**. The cooling tubes **12** define a primary cooling loop **14**. The cooling tubes **12** are retroactively thermally coupled to a main former or support shell or body **16** (e.g., main coil support) and, in an embodiment, may also be thermally coupled to a shield former (e.g., bucking former) or support shell or body **18** (e.g., shield coil support) that encompasses the main former **16**. The main former **16** and shield former **18** support or maintain the position of main MRI magnet coils **20** and shielding MRI magnet coils **22** (also referred to as bucking coils), respectively, in a manner heretofore known in the art. For example, the main magnet coils **20** may be shrink fit and bonded inside the main former **16**, which may be a cylindrical metal coil former, to thereby provide thermal contact therebetween. Likewise the shield magnet coils **22** may be shrink fit and bonded inside the shield former **18**, which may be a cylindrical metal coil former, to thereby provide thermal contact therebetween. Other types of coils may be provided, for example, epoxied coils. In an embodiment the main magnet coils **20** and the magnet shielding coils **22** may be formed from any material capable of producing a superconducting magnet, such as from niobium-titanium (NbTi), niobium-tin (Nb₃Sn) or magnesium-di-boride (MgB₂). As described in greater detail below, the cooling tubes **12** may be thermally coupled directly to the coils **20**, **22** on the formers **16**, **18**.

(26) As further illustrated in FIG. 2 (which illustrates an additional portion of the cooling system **10**), the various embodiments of the present invention may be implemented as part of an MRI superconducting magnet system **30**, wherein the cooling may be provided via a two stage cooling arrangement. For example, initial cooling may be provided via a helium bath and then subsequent cooling may be provided via circulating liquid helium through the cooling tubes.

(27) The coil formers **16**, **18** provide a cold mass support structure that maintain the position of or support the magnetic coils **20**, **22**, respectively (shielding magnet coils **22** not shown in FIG. 2). The coil formers **16**, **18** are made of a fiberglass composite material. In certain embodiments, the coil formers **16**, **18** may be formed from a thermally conductive material (e.g., aluminum). The cooling tubes **12**, which may be formed from any suitable metal (e.g., copper, stainless steel, aluminum, etc.) are in fluid communication with a liquid cryogen storage system **23** having one or more liquid cryogen storage tanks **24**. The cryogen storage system **23** contains a liquid cryogen used in the closed loop cooling system **10** to cool a cold mass **60**, including magnet coils **20**, **22**. In an embodiment, the cryogen is liquid helium. The fluid communication between the cooling tubes **12** and the liquid helium storage tank(s) **24** may be provided by one or more fluid passageways **26** (e.g., fluid tubes, conduits, etc.). Thus, the one or more storage tanks **24** provide the liquid helium that flows through the cooling tubes **12** to cool the magnet coils **20**, **22** by removing heat.

(28) Collectively, the assembly of magnet coils **20** and/or **22**, the coil formers/support structure **16** and/or **18** and the liquid cryogen storage tanks **24** form a cold mass **60**. As discussed in detail below, this cold mass **60** is cooled to target temperature. As used herein, “target temperature” means a cryogenic temperature sufficient to enable superconducting operation. In an embodiment, the target temperature is approximately 4 K. As used herein “cold mass” means any structure that is cooled to the target temperature during normal operation via the cooling loop **14**. In the illustrated embodiment, the cooling loop **14** contains no venting.

(29) As best illustrated in FIG. 2, in an embodiment, the cooling tubes **12** may be in fluid communication with a vapor return manifold or passageway **32** (e.g., tubes or conduits coupled to a recondenser **36**. receives helium vapor from the cooling tubes **12** that removes heat from the

magnet coils **20**, **22** and forms part of the closed loop cooling system.

(30) The liquid helium storage tanks **24** are in fluid communication with a cryorefrigerator **40** that includes the recondenser **36**. In various embodiments, the recondenser **36** operates to form a free convection circulation loop to cool the magnet coils **20**, **22** and coil support shells **16**, **18** to a cryogenic temperature, as well as fills the liquid helium storage tanks **24** with liquid helium via one or more passageways **44**. The cryorefrigerator **40** may be a coldhead or other suitable cryocooler. The cryorefrigerator **40** also includes a motor **50**.

(31) As illustrated in FIG. 2, the cryorefrigerator **40** in various embodiments includes the recondenser **36** at a lower end of the cryorefrigerator **40** that recondenses boiled off helium gas received from the vapor return manifold/passageway **32** in parallel with the helium gas storage system **34**. The recondenser **36** allows for of liquid helium to the liquid helium reservoir **24**.

(32) The magnet coils **20**, which in various embodiments are molded coils, form a main superconducting magnet **52** that is controlled during operation of the MRI system as is known in the art to acquire MRI image data. Additionally, during operation of the MRI system, liquid helium travelling through the thermally coupled cooling tubes **12** cools the superconducting magnet **52**. The superconducting magnet **52** may be cooled, for example, to a superconducting temperature, such as 4.2 Kelvin (K). A portion of the cooling process may include the recondensing of boiled off helium gas to liquid by the recondenser **36** and returned to the liquid helium tank **24**, as well as cooling of the boiled off helium.

(33) In addition to cooling via the circulation of liquid helium through cooling tubes that are thermally connected to the magnet, cooling may be provided by immersing the superconducting magnet coils in a bath of liquid helium. In various embodiments, once cooled to operating temperature, the magnet coils may be cooled by thermal conduction and/or by thermosiphoning cooling. As will be readily appreciated, however, whether cooling is effected through the circulation of a liquid cryogen through a cooling loop that is in thermal communication with the magnet, or through the immersion of magnet coils in a bath of liquid cryogen, the coil formers/support structure, magnet coils and/or liquid cryogen reservoir form a cold mass **60** having a temperature of approximately 4.2 K, which provides for superconducting operations.

(34) As depicted, the coil support shell **20**, the cooling tubes **12**, and the liquid helium storage system **23** are disposed within a cryogenic vessel **48** (e.g., helium vessel). The cooling tubes **12** may be thermally coupled to the cryogenic vessel **48**. The cryogenic vessel **48** may be partially filled with liquid helium to cool the main superconducting magnet **52** as noted above. The cryogenic vessel **48** may disposed within a thermal shield **54** (e.g., a thermally isolating radiation shield). The cryorefrigerator **40** (including the motor **50** and the recondenser) **36**) are located outside both the cryogenic vessel **48** and the thermal shield **54**. The temperature at the thermal shield **54** may be approximately 50 K.

(35) A pump **72** (e.g., vacuum or roughing pump) is connected to the cryogenic vessel **48** to allow helium gas to be pumped directly out of the cryogenic vessel **48**. For example, during pump down and cooling down, helium gas may be pumped out to reduce the pressure (e.g., as to low as tens of millitorr) and, thus, the temperature within the cryogenic vessel **48**. The pump **72** may be coupled to a helium gas storage unit **74**. A controller **76** (e.g., having processing circuitry and memory circuitry) is connected to the pump **72** to control the operation of the pumping, for example, to control the level of pressure within the cryogenic vessel **48** during ramping. For example, the controller **76** may be connected to a temperature sensing device (not shown) or pressure sensing device (not shown) that measure the temperature of and pressure in the cryogenic vessel **48**, respectively. Thus, the controller **76** may control the pump **72** such that reduced pressure in the helium vessel is provided during ramping.

(36) As noted above, the cooling system **10** for the superconducting magnet **52** is retrofitted to form a sealed low cryogenic system. Besides providing the cooling tubes **10**, the retrofit includes providing for feedthrough configurations that eliminate vibration transmission from the coldhead

and that compensate for thermal expansion differentials between the materials involved. This may include forming a vacuum seal **78** between any tube or conduit that extends between the recondenser **36** and the liquid storage system **23** or any other location within the cryogenic vessel **48** (e.g., passageways **26**, **32**). providing for feedthrough configurations (e.g., for tubes through the sealed helium vessel) that eliminate vibration transmission from the coldhead and that compensate for thermal expansion differentials between the materials involved. Other tubes or conduits coupled to the cryogenic vessel **48** may be vacuum sealed where they pass thru the cryogenic vessel **48**. Adding the vacuum seals makes the cryogenic vessel **48** a sealed cryogenic vessel **48**.

(37) The superconducting magnet **52** is electrically coupled to a superconducting switch **53**. Retrofitting the cooling system **10** includes replacing the original main superconducting switch with the superconducting switch **53** that is configured to operate at a temperature of 80 Kelvin or less. As described in greater detail below, the superconducting switch **53** is both located within and thermally isolated from the helium vessel **48**. The superconducting switch **53** is configured to switch between a resistive mode and a superconducting mode. In addition, the superconducting switch **53** is thermally coupled to a respective tube **10** of the cooling tubes **10**. Further, a high temperature superconducting current leads may be provided and coupled to the superconducting switch **53**.

(38) FIG. **3** is a cross-sectional view of a cooling tube **13** of the cooling system **10** in FIG. **1**. As noted above, the cooling tube **12** may be formed from any suitable metal (e.g., copper, stainless steel, aluminum, etc.). As depicted, the cooling tube **12** is configured for thermocoupling to a coil of the superconducting magnet. In particular, a thermoconductive sheet **80** made of metal (e.g., copper) is wrapped or disposed about the cooling tube **12**. The thermoconductive sheet **80** is configured maximizes the heat transfer surface for the cooling tube **12**. A thermoconductive plate **82** (e.g., copper plate) is coupled to and flanked by ends **84** of the thermoconductive sheet **80** via a fastener **85** (e.g., bolt) that passes through the ends **84** and the thermoconductive plate **82**. The thermoconductive plate **82** may also be soldered to the ends **84** of the thermoconductive sheet **82**. The thermoconductive plate **82** acts as a thermoconductive conduit between the cooling tube **12** and the coil of the superconducting magnet. A single cooling tube **12** may be coupled to multiple coils of the superconducting magnet. Thermocoupling directly of the coils to the superconducting magnet helps form a sealed cryogenic system since the liquid helium is kept inside the cooling tubes **12**.

(39) FIG. **4** is a schematic diagram illustrating a process for thermocoupling a coil **86** (e.g., coil **20** or **22** in FIG. **1**) of a superconducting magnet to the cooling tube **12** in FIG. **3** (e.g., using metal wire overwrap). The coil **86** is configured to be disposed on a former made of fiberglass composite. A flexible thermoconductive strap or braid **88** (e.g., made of a metal such as copper) is disposed or laid on the top of the coil **86**. Then a thermoconductive wire overwrap **90** (e.g., made of a metal such as copper) is wound directly on top (and around) the coil **86** to compress the flexible thermoconductive strap or braid **88** against the coil **86** to form a better thermal contact. In certain embodiments, a thermal grease can be disposed between the thermoconductive strap or braid **88** and the coil **86** to increase efficiency. The thermoconductive strap or braid **88** and the thermoconductive wire overwrap **90** from a thermocoupling assembly **92** to directly thermocouple the coil **86** to the cooling tube **12**. Ends **94** of the thermoconductive strap or braid **88** are folded up and connected to the thermoconductive plate **82** coupled to the cooling tube **12**. This configuration takes advantage of the very high heat transfer along the hoop direction to minimize the surface area of the thermoconductive strap or braid **88**. Brazing or soldering between the metal (e.g., copper) parts and the cooling tube **12** can be utilized to increase the heat transfer efficiency.

(40) FIG. **5** is a schematic diagram illustrating a process for thermocoupling the coil **86** (e.g., coil **20** or **22** in FIG. **1**) of a superconducting magnet to the cooling tube **12** in FIG. **3** (e.g., using conforming metal plate). The coil **86** is configured to be disposed on a former made of fiberglass composite. A flexible thermoconductive strap or braid **88** (e.g., made of a metal such as copper) is

disposed or laid on the top of the coil **86**. Then, a thermoconductive conforming plate **96** (e.g., made of a metal such as copper) is fastened (e.g., bolted) to flanges of the former (see FIG. **6**) on top of a portion of the thermoconductive strap or braid **88** to compress the flexible thermoconductive strap or braid **88** against the coil **86** to form a better thermal contact. In certain embodiments, a thermal grease can be disposed between the thermoconductive strap or braid **88** and the coil **86** to increase efficiency. The thermoconductive strap or braid **88** and the thermoconductive conforming plate **96** from a thermocoupling assembly **98** to directly thermocouple the coil **86** to the cooling tube **12**. Ends **94** of the thermoconductive strap or braid **88** are folded up and connected to the thermoconductive plate **82** coupled to the cooling tube **12**. This configuration takes advantage of the very high heat transfer along the hoop direction to minimize the surface area of the thermoconductive strap or braid **88**. Brazing or soldering between the metal (e.g., copper) parts and the cooling tube **12** can be utilized to increase the heat transfer efficiency. (41) FIG. **6** is a cross-sectional view of the cooling tube thermally coupled directly to the coil **86** in FIG. **5**. As depicted, the thermoconductive conforming plate **96** (e.g., made of a metal such as copper) is fastened to flanges **100** of the former **102** (e.g., coil former **16** or **18** in FIG. **1**) on top of a portion of the thermoconductive strap or braid **88** to compress the flexible thermoconductive strap or braid **88** against the coil **86** to form a better thermal contact.

(42) FIG. **7** is a liquid helium network **104** for cooling a superconducting magnet of an MRI machine. Annulus **106** represents an entry point through the thermal shield (e.g., thermal shield **54** in FIG. **2**). Annulus **108** represents an entry point through the cryogenic vessel or helium vessel (e.g., cryogenic vessel **48** in FIG. **2**). Annulus **110** represents the entry point or main penetration point for liquid helium to be provided to fill cryogenic vessel. The main penetration point may be heat sunk to first stage on bellows. Annuli **112** represent entry points through the bucking former (e.g., coil former **18** in FIG. **1**). Dashed line **114** represents a level of liquid helium in the cryogenic vessel. The sealed cryogenic system liquid helium storage tanks **24** are located within the cryogenic vessel between the main and bucking formers. Space availability can dictate that the helium gas be stored under very high pressures. The gas will slowly turn into vapor and liquid as the magnet cools down from room temperatures. The recondenser **36** is located outside both the thermal shield and the cryogenic vessel. Each liquid helium storage tank **24** may be coupled to the recondenser **36** via respective tube **115** that extends from inside the cryogenic vessel to outside both the cryogenic vessel and the thermal shield.

(43) As depicted, a network **116** of the cooling tubes **12** is disposed throughout the cryogenic vessel forming a sealed cryogenic system. The cooling tubes **12** may be thermally coupled to the coils and the cryogenic vessel. Tubes **118** represent cooling tubes **12** that are utilized for cooling the bucking coils (e.g., coils **22** in FIG. **1**). Tubes **120** represent cooling tubes **12** that are utilized for cooling the main coils (e.g., coils **20** in FIG. **1**). Tube **122** represents a cooling tube **12** utilized to cool the superconducting switch (e.g., superconducting switch **53** in FIG. **2**). The liquid helium within the tubes may be converted into vapor upon heat transfer from the various coils and other components. The portion of the cooling tubes **12** may be fluidly coupled to a manifold **124** that combines the flow into a single tube for returning the vapor liquid helium to the recondenser **36**.

(44) Some tubes or conduits (e.g., the liquid/vapor line between the tanks **24** and the recondenser) may require passing through vacuum boundaries. These tubes or conduits would need vacuum feedthroughs (e.g. that form vacuum seals) eliminate vibration transmission from the coldhead and that compensate for thermal expansion differentials between the materials involved. FIGS. **8** and **9** describe different examples of vacuum feedthroughs to utilize to create the sealed low cryogenic system.

(45) FIG. **8** is a schematic diagram of a vacuum feedthrough between the recondenser **36** and the tank **24** (e.g., liquid helium storage tank) of the cooling system **10** in FIG. **2** (e.g., bellows). As depicted, the recondenser **36** is located outside the cryogenic vessel **48** and the tank **24** is disposed within the cryogenic vessel **48** (see FIG. **2**). The recondenser **36** is coupled to the tank **24** via a tube

126 (e.g., passageway **44** in FIG. 2). The tube **126** extends through an opening **128** in the cryogenic vessel **48**. A bellows **130** is located at the opening **128** to form a vacuum feedthrough (e.g., to form a vacuum seal or boundary **132**) for the tube **126** to extend through the opening **128**. The bellows **130** is disposed along the tube **126** adjacent the opening **128**. The tube **126** is a rigid tube. Also, the tube **126** includes a near horizontal strain relief curve (as indicated by reference numeral **134**) adjacent the tank **24**.

(46) FIG. 9 is a schematic diagram of a vacuum feedthrough between the recondenser **36** and the tank **24** (e.g., bi-metallic interface) of the cooling system **10** in FIG. 2 (e.g., bellows). As depicted, the recondenser **36** is located outside the cryogenic vessel **48** and the tank **24** is disposed within the cryogenic vessel **48** (see FIG. 2). The recondenser **36** is coupled to the tank **24** via a tube **136** (e.g., passageway **44** in FIG. 2). The tube **136** extends through an opening **128** in the cryogenic vessel **48**. A bi-metallic interface **138** is located at the opening **128** to form a vacuum feedthrough (e.g., to form a vacuum seal or boundary **140**) for the tube **136** to extend through the opening **128**. The bi-metallic interface **138** is welded to the cryogenic vessel **48**. The tube **136** extends through the bi-metallic interface **138** so that the bi-metallic interface **138** is disposed about the tube **136**. The tube **136** is a high pressure flexible tube. As depicted in FIG. 10, the bi-metallic interface **138** includes an inner metal layer **141** that interfaces with an outer surface of the tube **136** and an outer metal layer **142** that interfaces with the cryogenic vessel **48**, and the inner metal layer **141** is made of a different metallic material than the outer metal layer **142**. For example, the inner metal layer **141** may be made of stainless steel and the outer metal layer **142** may be made of aluminum.

(47) FIG. 11 is a schematic diagram of the superconducting switch **53**. The superconducting switch **53** is configured to operate at a temperature of 80 Kelvin or less. In particular, the superconducting switch **53** is configured for low cryogen operation. The superconducting switch **53** is both located within (e.g., see FIG. 2) and thermally isolated from the cryogenic vessel. The superconducting switch **53** is configured to switch between a resistive mode and a superconducting mode.

(48) As depicted, the superconducting switch **53** includes a body **144**. A thermoconductive metallic sheet **146** (e.g., copper sheet) is disposed about the body **144**. The thermoconductive metallic sheet **146** enables switch temperature control. The superconducting switch **53** includes a blanket or cryogenic insulation **148** (e.g., Cryogel® Z) disposed about the thermoconductive metallic sheet **146**. The blanket insulation **148** is configured to thermally isolate the superconducting switch **53** from the cryogenic vessel. The blanket insulation **148** functions well at low vacuum levels as disclosed herein. The superconducting switch **53** also includes a thermoconductive metallic sheet **150** (e.g., cooper sheet) disposed about the blanket insulation **148**. The superconducting switch **53** is thermally coupled to the cooling tube **12**. In particular, the thermoconductive metallic sheet **150** is coupled via thermal short **152**. The superconducting switch **53** is also coupled to the cooling tube via a calibrated thermal shunt **154**. The blanket insulation **148** and the thermoconductive metallic sheet **150** are configured to keep convection from being generated by the superconducting switch **53** during ramping of the superconducting magnet. The thermal isolation of the superconducting switch **53** enables the superconducting switch **53** to go to higher temperature under driving mode and thus low boil-off during ramping.

(49) To retroactively create the sealed low cryogenic system power leads including a high temperature superconducting wire need to be coupled to the superconducting switch. FIG. 12 is a schematic diagram of a current or power leads **155** coupled to the superconducting switch **53**. As depicted, the superconducting switch **53** is located within the cryogenic vessel **48**. A first heat station **156** (e.g., pad) is located at the thermal shield **54** (which is approximately 50 K) and a second heat station **158** (e.g., pad) is located at the cryogenic vessel **48** (e.g., at approximately 4 K). The first heat station **156** and the second heat station **158** are configured to provide thermal contacts and electrical isolation for the current or power leads **155** at the thermal shield **54** and the cryogenic vessel **48**, respectively. The superconducting current or power leads **155** include a high temperature superconducting wire **160** (which functions as the ramp lead) located outside the

cryogenic vessel **48**. A first end **162** (e.g., cold end) of the high temperature superconducting wire **160** is coupled to the first heat station **156** at the thermal shield **54**. A second end **164** (e.g., warm end) of the high temperature superconducting wire **160** is coupled to the second heat station **158** at the cryogenic vessel **48**. Both ends **162**, **164** of the high temperature superconducting wire **160** may be coated with alumina ceramic for further electrical isolation. The superconducting current or power leads **155** also includes a low temperature superconducting wire **166**. The low temperature superconducting wire **166** has a first end **168** coupled to the second heat station **158** outside the cryogenic vessel **48**. A second end **170** of the low temperature superconducting wire **166** is coupled to the superconducting switch **53** within the cryogenic vessel **48**. The low temperature superconducting wire **166** passes from outside the cryogenic vessel **48** through an electrically isolated feedthrough **172** into the cryogenic vessel **48**.

(50) FIG. **13** is a flowchart of a method **174** for cooling a superconducting magnet for a magnetic resonance imaging system. The method **174** includes filling a sealed cryogenic vessel with liquid helium until a first desired temperature is reached for cooling the superconducting magnet having a plurality of coils disposed about a body of a coil support structure, wherein the superconducting magnet is disposed within the sealed cryogenic vessel (block **176**). The method also includes ceasing filling the sealed cryogenic vessel with the liquid helium upon reaching the first desired temperature (block **178**). The method further includes utilizing a roughing vacuum pump coupled to the sealed cryogenic vessel to evacuate excess helium gas after ceasing filling the sealed cryogenic vessel with the liquid helium (block **180**). The method even further includes activating a sealed cryogenic system including a network of tubes disposed within the sealed cryogenic vessel by flowing a cryogen through the network of tubes to cool the superconducting magnet to a second desired temperature, wherein the second desired temperature is lower than the first desired temperature (block **182**). The pressure in the sealed cryogenic vessel after evacuating the excess helium gas generates internal convection to ensure a uniform temperature distribution between the plurality of coils and the sealed cryogenic vessel. The pressure of the leftover gas after cooldown and pump can be in the tens of millitorr. This in conjunction with the internal convection enables the uniform temperature distribution. The method **174** enables the retrofitted sealed low cryogen system to cool the superconducting magnet quicker than a typical sealed low cryogen system. Indeed, the method **174** enables the retrofitted sealed low cryogen system to cool the superconducting magnet similar to helium bath cooled system. In certain embodiments, the superconducting magnet may be directly cooled to the second desired temperature only utilizing the flow of helium through the network of tubes (i.e., without initially utilizing a helium bath for cooling the superconducting magnet).

(51) FIG. **14** illustrates a schematic block diagram of an example MR system **200** having the retrofitted sealed cryogen system disclosed herein. MR system **200** is used to obtain images or for spectroscopy applications of a subject. In the example embodiment, MR system **200** includes a magnet assembly **202** that includes a magnet **204**. In some embodiments, magnet **204** is a superconducting magnet formed from a plurality of magnetic coils wound around a magnetic coil support or a coil former. Magnet **204** is configured to generate a polarizing magnetic field. Magnet assembly **202** may include a cryostat or cryogenic vessel **205** that surrounds magnet **204**. Cryostat vessel **205** is typically filled with a cryogenic fluid or cryogen which is used to cool the superconducting coils into an extremely low temperature, e.g., 4 K, such that electric current continues to flow through the superconducting coils without electrical resistance to maintain a uniform and static magnetic field after a power supply is disconnected. Cryogen may be helium, hydrogen, neon, nitrogen, or any combination thereof, in a liquid form, a gaseous form, or a combination of liquid and gaseous form. Helium is described as an example cryogen. Other cryogen may be used in superconducting magnet assemblies described herein.

(52) In the example embodiment, magnet assembly **202** may also include a thermal shield assembly **206** that enclose cryostat vessel **205** and magnet **204** therein. In one embodiment, thermal shield

assembly **206** may include an inner thermal shield member **208** and an outer thermal shield member **210**. Inner thermal shield member **208** is generally cylindrical in shape and is radially placed inside of magnet **204**. Inner thermal shield member **208** is configured to prevent heat being radiated from a warm region where the subject is placed to a cold region where magnet **204** is placed. Outer thermal shield member **210** is arranged concentrically with respect to inner thermal shield member **208**. Outer thermal shield member **210** may also have a generally cylindrical shape and is radially placed outside of magnet **204**. Outer thermal shield member **210** is configured to prevent heat being radiated from environment into magnet **204**. Thermal shield assembly **206** is made from metal materials, such as aluminum.

(53) In the example embodiment, MR system **200** also includes a gradient coil assembly **212** placed inside of inner thermal shield member **208**. Gradient coil assembly **212** is configured to selectively impose one or more gradient magnetic fields along one or more axes, such as x, y, or z axes. MR system **200** also includes radio frequency (RF) coil **214**. RF coil **214** may be a transmitter coil, which is configured to transmit RF pulses. RF coil **214** may be a receiver coil, which is configured to detect MR signals from the subject. RF coil **214** may be a transmit and receive coil that transmits and also detect MR signals. Magnet assembly **202**, gradient coil assembly **212**, and body RF coil **214** are collectively referred to as a scanner assembly **216**, because scanner assembly **216** forms into one unit and is in a scanner room. Scanner assembly **216** has a bore **217**, where the subject is positioned during scanning. Scanner assembly **216** shown in FIG. 1 is a closed bore system, where the bore is cylindrical. Scanner assembly **216** may be magnet assemblies of other designs, such as an open-bore system, a dipolar electromagnet configuration, or a Hallbach configuration.

(54) In the example embodiment, MR system **200** also includes a controller **218**, a magnetic field control **220**, a gradient field control **222**, a memory **224**, a display device **226**, a transmit/receive (T/R) switch **228**, an RF transmitter **230**, and a receiver **232**. In operation, a subject is placed in bore **217** on a suitable support, for example, a motorized table (not shown) or other patient table. Magnet **204** produces a uniform and static magnetic field $B_{sub.0}$ across bore **217**. Strength and homogeneity of the magnet field B_0 in bore **217** and correspondingly in patient is controlled by controller **218** via magnetic field control **220**, which also controls a supply of energized current to magnet **204**. Gradient coil assembly **212** is energized by gradient field control **222** and is also controlled by controller **218**, so that one or more gradient magnetic fields are imposed on the magnetic field $B_{sub.0}$. RF coil **214** and a receive coil, if provided, are selectively interconnected to one of RF transmitter **230** or receiver **232**, respectively, by T/R switch **228**. RF transmitter **230** and T/R switch **228** are controlled by controller **218** such that RF field pulses or signals are generated by RF transmitter **230** and are selectively applied to the subject for excitation of magnetic resonance in the subject.

(55) In the example embodiment, following application of the RF pulses, T/R switch **228** is again actuated to decouple RF transmit coil **214** from RF transmitter **230**. The detected MR signals are in turn communicated to controller **218** which may organize the MR signals in a particular format for storage in memory **224**. Controller **218** includes a processor **234** that controls the processing of the MR signals to produce signals representative of an image of the patient, which are transmitted to display device **226** to provide a visual display of the image.

(56) Technical effects of the disclosed subject matter include providing for converting existing bath cooled magnets (especially those with fiberglass composite structures) to sealed low cryogen systems while keeping the support structure, coil configuration, and cryostat (e.g., helium vessel) nearly intact, while replacing a main switch of the superconducting magnet with a switch configured for low cryogen operation and adding superconducting current leads. In addition, technical effects of the disclosed subject matter include providing for a liquid helium network that allows the cooling of the sealed helium vessel directly. Further, the technical effects of the disclosed subject matter include providing for feedthrough configurations (e.g., for tubes through

the sealed helium vessel) that eliminate vibration transmission from the coldhead and that compensate for thermal expansion differentials between the materials involved. Even further, the technical effects of the disclosed subject matter include enabling a faster cooldown for the superconducting magnet disposed in the sealed cryogenic vessel (e.g., from 1 to 3 weeks to 1 to 3 days) which is equivalent to cooldown for a bath cooled magnet. Still further, technical effects of the disclosed subject matter includes reducing helium consumption (e.g., due to eliminating helium addition during a lifetime of the magnet) and, thus, the costs associated with consumption of helium. Yet further, technical effects of the disclosed subject matter include eliminating the need for a quench vent pipe.

(57) The techniques presented and claimed herein are referenced and applied to material objects and concrete examples of a practical nature that demonstrably improve the present technical field and, as such, are not abstract, intangible or purely theoretical. Further, if any claims appended to the end of this specification contain one or more elements designated as “means for [perform]ing [a function] . . . ” or “step for [perform]ing [a function] . . . ”, it is intended that such elements are to be interpreted under 35 U.S.C. 112(f). However, for any claims containing elements designated in any other manner, it is intended that such elements are not to be interpreted under 35 U.S.C. 112(f).

(58) This written description uses examples to disclose the present subject matter, including the best mode, and also to enable any person skilled in the art to practice the subject matter, including making and using any devices or systems and performing any incorporated methods. The patentable scope of the subject matter is defined by the claims, and may include other examples that occur to those skilled in the art. Such other examples are intended to be within the scope of the claims if they have structural elements that do not differ from the literal language of the claims, or if they include equivalent structural elements with insubstantial differences from the literal languages of the claims.

Claims

1. A superconducting magnet system for a magnetic resonance imaging system, comprising: a coil support structure having a body; and a superconducting magnet having a plurality of coils disposed about the body of the coil support structure; and a helium vessel encompassing the coil support structure and the superconducting magnet; and a cooling system comprising: a network of tubes disposed within the helium vessel and configured to transport helium within, wherein the network of tubes is retroactively thermally coupled to the plurality of coils; a liquid helium storage system located within the helium vessel; a helium recondenser located outside the helium vessel to recondense vapor helium to liquid helium; and one or more tubes coupling the helium storage system to the helium recondenser, wherein where the one or more tubes pass through the helium vessel is retroactively vacuum sealed to make the helium vessel a sealed helium vessel, wherein the cooling system is configured to keep a uniform temperature between the sealed helium vessel and the plurality of coils; and wherein each coil of the plurality of coils is thermally coupled to a respective tube of the network of tubes via a thermocoupling assembly, wherein the thermocoupling assembly comprises a flat thermoconductive metal strap disposed on a respective coil, a thermoconductive metallic wire overwrap wound about both a portion of the flat thermoconductive metallic strap and the respective coil, and ends of the flat thermoconductive metal strap are coupled to a thermoconductive metallic plate coupled to the respective tube.
2. The superconducting magnet system of claim 1, further comprising a respective bellows located where each respective tube of the one or more tubes pass through the helium vessel, wherein each respective bellows forms a vacuum seal and is coupled to the respective tube of the one or more tubes, and the one or more tubes are rigid.
3. The superconducting magnet system of claim 1, further comprising a respective bi-metallic interface disposed about each tube of the one or more tubes where each respective tube of the one

or more tubes passes through the helium vessel, wherein each respective bi-metallic interface forms a vacuum seal, and the one or more tubes are flexible.

4. The superconducting magnet system of claim 3, wherein each respective bi-metallic interface comprises an inner metal layer that interfaces with the respective tube and an outer metal layer that interfaces with the helium vessel, and the inner metal layer is made of a different metallic material than the outer metal layer.

5. The superconducting magnet system of claim 1, wherein the thermocoupling assembly comprises a thermoconductive metallic sheet wrapped around the respective tube, and the thermoconductive metallic plate is coupled to the thermoconductive metallic sheet.

6. The superconducting magnet system of claim 1, further comprising a superconducting switch configured to operate at a temperature of 80 Kelvin or less, wherein the superconducting switch is both located within and thermally isolated from the helium vessel, the superconducting switch is configured to switch between a resistive mode and a superconducting mode, and the superconducting switch is thermally coupled to a respective tube of the network of tubes.

7. The superconducting magnet system of claim 6, further comprising power leads coupled to the superconducting switch, wherein the power leads comprise a high temperature superconducting wire coupled to a heat sink located outside the helium vessel, wherein an end of the high temperature superconducting wire coupled to the heat sink is electrically isolated, and wherein the power leads comprise a low temperature superconducting wire having a first end coupled to the heat sink and a second end coupled to the superconducting switch, and the low temperature superconducting wire passes from outside the helium vessel through an electrically isolated feedthrough into the helium vessel.

8. The superconducting magnet system of claim 1, wherein the body of the coil support structure is made of fiberglass composite.

9. A method for cooling a superconducting magnet for a magnetic resonance imaging system, comprising: filling a sealed cryogenic vessel with liquid helium until a first desired temperature is reached for cooling the superconducting magnet having a plurality of coils disposed about a body of a coil support structure, wherein the superconducting magnet is disposed within the sealed cryogenic vessel; ceasing filling the sealed cryogenic vessel with the liquid helium upon reaching the first desired temperature; utilizing a roughing vacuum pump coupled to the sealed cryogenic vessel to evacuate excess helium gas after ceasing filling the sealed cryogenic vessel with the liquid helium; and activating a sealed cryogenic system comprising a network of tubes disposed within the sealed cryogenic vessel by flowing a cryogen through the network of tubes to cool the superconducting magnet to a second desired temperature, wherein the second desired temperature is lower than the first desired temperature.

10. The method of claim 9, wherein pressure in the sealed cryogenic vessel after evacuating the excess helium gas generates internal convection to ensure a uniform temperature distribution between the plurality of coils and the sealed cryogenic vessel.

11. The method of claim 9, wherein each coil of the plurality of coils is retroactively thermally coupled to a respective tube of the network of tubes via a thermocoupling assembly, wherein the thermocoupling assembly comprises a flat thermoconductive metal strap disposed on a respective coil, a thermoconductive metallic wire overwrap wound about both a portion of the flat thermoconductive metallic strap and the respective coil, and ends of the flat thermoconductive metal strap are coupled to a thermoconductive metallic plate coupled to the respective tube.

12. The method of claim 9, wherein the magnetic resonance imaging system comprises a cooling system for the superconducting magnet, wherein the cooling system comprises a liquid helium storage system located within the sealed cryogenic vessel, a helium recondenser located outside the cryogenic vessel to recondense vapor helium to liquid helium, and one or more tubes coupling the helium storage system to the helium recondenser, wherein where the one or more tubes pass through the sealed cryogenic vessel is retroactively vacuum sealed to a make a cryogenic vessel the

sealed cryogenic vessel, and wherein the cooling system is configured to keep a uniform temperature between the sealed helium vessel and the plurality of coils.

13. The method of claim 12, wherein the cooling system comprises a respective bi-metallic interface disposed about each tube of the one or more tubes where each respective tube of the one or more tubes passes through the sealed cryogenic vessel, wherein each respective bi-metallic interface forms a vacuum seal, and the one or more tubes are flexible.

14. The method of claim 9, wherein the magnetic resonance imaging system comprises a cooling system for the superconducting magnet, wherein the cooling system comprises a superconducting switch configured to operate at a temperature of 80 Kelvin or less, wherein the superconducting switch is both located within and thermally isolated from the sealed cryogenic vessel, the superconducting switch is configured to switch between a resistive mode and a superconducting mode, and the superconducting switch is thermally coupled to a respective tube of the network of tubes.

15. The method of claim 9, wherein the cooling system comprises power leads coupled to the superconducting switch, wherein the power leads comprise a high temperature superconducting wire coupled to a heat sink located outside the sealed cryogenic vessel, wherein an end of the high temperature superconducting wire coupled to the heat sink is electrically isolated, and wherein the power leads comprise a low temperature superconducting wire having a first end coupled to the heat sink and a second end coupled to the superconducting switch, and the low temperature superconducting wire passes from outside the sealed cryogenic vessel through an electrically isolated feedthrough into the sealed cryogenic vessel.

16. A magnetic resonance imaging (MRI) system, comprising: a coil support structure having a body, wherein the body of the coil support structure is made of fiberglass composite; and a superconducting magnet having a plurality of coils disposed about the body of the coil support structure; and a helium vessel encompassing the coil support structure and the superconducting magnet; and a cooling system comprising: a network of tubes disposed within the helium vessel and configured to transport helium within, wherein the network of tubes is retroactively thermally coupled to the plurality of coils; a liquid helium storage system located within the helium vessel; a helium recondenser located outside the helium vessel to recondense vapor helium to liquid helium; and one or more tubes coupling the helium storage system to the helium recondenser, wherein where the one or more tubes pass through the helium vessel is retroactively vacuum sealed to make the helium vessel a sealed helium vessel, wherein the cooling system is configured to initially cool the superconducting magnet to a first desired temperature via filling the sealed helium vessel with liquid helium and to subsequently cool the superconducting magnet to a second desired temperature via the helium flowing through the network of tubes, wherein the second desired temperature is lower than the first desired temperature; and wherein each coil of the plurality of coils is thermally coupled to a respective tube of the network of tubes via a thermocoupling assembly, wherein the thermocoupling assembly comprises a flat thermoconductive metallic strap disposed on a respective coil, a thermoconductive metallic plate disposed about a portion of the flat thermoconductive metallic strap, and ends of the flat thermoconductive metallic strap are coupled to an additional thermoconductive metallic plate coupled to the respective tube.

17. A superconducting magnet system for a magnetic resonance imaging system, comprising: a coil support structure having a body; and a superconducting magnet having a plurality of coils disposed about the body of the coil support structure; and a helium vessel encompassing the coil support structure and the superconducting magnet; and a cooling system comprising: a network of tubes disposed within the helium vessel and configured to transport helium within, wherein the network of tubes is retroactively thermally coupled to the plurality of coils; a liquid helium storage system located within the helium vessel; a helium recondenser located outside the helium vessel to recondense vapor helium to liquid helium; one or more tubes coupling the helium storage system to the helium recondenser, wherein where the one or more tubes pass through the helium vessel is

retroactively vacuum sealed to make the helium vessel a sealed helium vessel, wherein the cooling system is configured to keep a uniform temperature between the sealed helium vessel and the plurality of coils; and a superconducting switch configured to operate at a temperature of 80 Kelvin or less, wherein the superconducting switch is both located within and thermally isolated from the helium vessel, the superconducting switch is configured to switch between a resistive mode and a superconducting mode, and the superconducting switch is thermally coupled to a respective tube of the network of tubes; and wherein the superconducting switch comprises a blanket insulation disposed about a body of the switch and configured to thermally isolate the superconducting switch from the helium vessel, the superconducting switch comprises a thermoconductive metallic sheet disposed about the blanket insulation and thermally connected to the respective tube, and wherein the blanket insulation and the thermoconductive metallic sheet are configured to keep convection from being generated by the superconducting switch during ramping of the superconducting magnet.
